CSM
Fall 2026

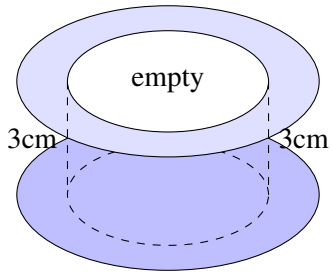
EE 64
Worksheet 6

1. Yes Cap

Learning Goal: Learn of the fundamentals of capacitors and equivalent capacitance.

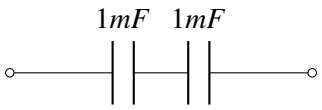
Capacitor Sword Man is one of the most well known brand for selling swords in the galaxy. Recently, they have came to Earth on a field trip, and picked up some almost broken capacitors and resistors form the floor of Cory Hall to experiment new lightsaber switches.
Cap Sword Man (CSM) has now appeared in your dream to ask you about capacitors.

(a) Let us first look at this structure:



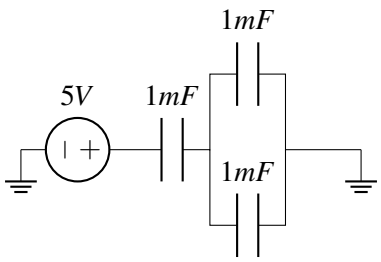
The colored rings and circles are conductor plates, each with an outer radius of 4cm. Meanwhile, the top conductor plate is a ring whose inner radius is 3cm.
The permittivity of the material they put between the conductor ring and the conductor plate is $10F/m$. The vertical distance between the ring and the plate is 3 cm. What is the capacitance of this capacitor CSM just built?

(b) CSM has decided to just use the capacitors they picked up from Cory Floor instead, but they only picked up $1mF$ capacitors. They now use the knowledge of resistors and attempt to see if the following combination produces a capacitor whose equivalent capacitance is different from $1mF$:



Calculate the equivalent capacitance, and **in addition**, derive the capacitance formula for capacitors in series.

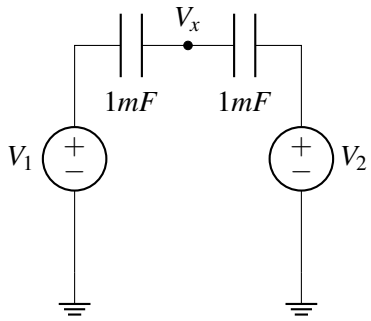
(c) CSM is satisfied with your answer. Then, CSM continues to use the knowledge of resistors, and attempt to see the equivalent capacitance and potential energy that the following configuration produces:



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Calculate the energy of capacitors for them.

(d) CSM notices from your lecture notes that there is something called "superposition". He also noticed that he only has two voltage sources, both of which are not of the supply he wants; however, its difference is.
Derive the voltage at the marked node (V_x) on the circuit, and verify whether superposition works to solve it:



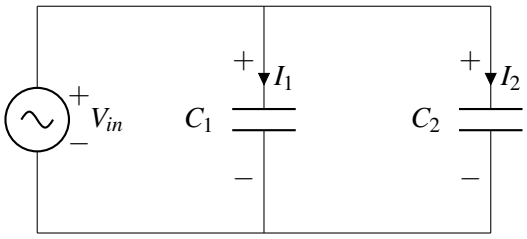
(e) Now, CSM replaces the $1mF$ capacitor at the right of the above architecture with a custom capacitor that is combined with the lightsaber switch button. Therefore, the capacitor at the right now has a variable capacitance such that for d being the distance that button is pressed into,

$$C(d) = \frac{10^{-3}}{0.1 - d} F$$

You also have a voltmeter which can return the voltage it reads to a computer. Once the computer chip receive some voltage lower than a specific value, it will turn the switch on and activate the lightsaber. Now, given the above architecture, find ways to fit the voltmeter and computer chip into the circuit architecture to achieve the above demands, and calculate V_x as a function of d.

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2. Parallel Capacitors



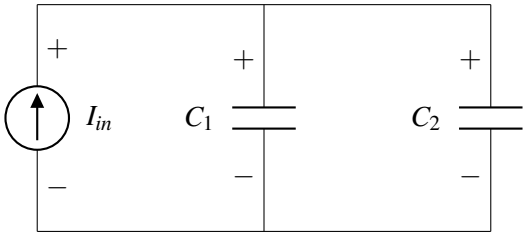
Suppose $V_{in}(t) = 2sin(x)$ and $C_1 = 2F, C_2 = 3F$

(a) Find the current I_1 after 2π seconds

(b) Find the current I_2 after 3π seconds

(c) Derive an expression for the equivalent capacitance between C_1 and C_2 and evaluate the numerical value

(d) Now, suppose our voltage source was replaced with a current source $i(t) = 3A$. as shown in the figure below



What is the voltage across capacitor C_1 after 2 seconds. Assume both capacitors are initially uncharged

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